README

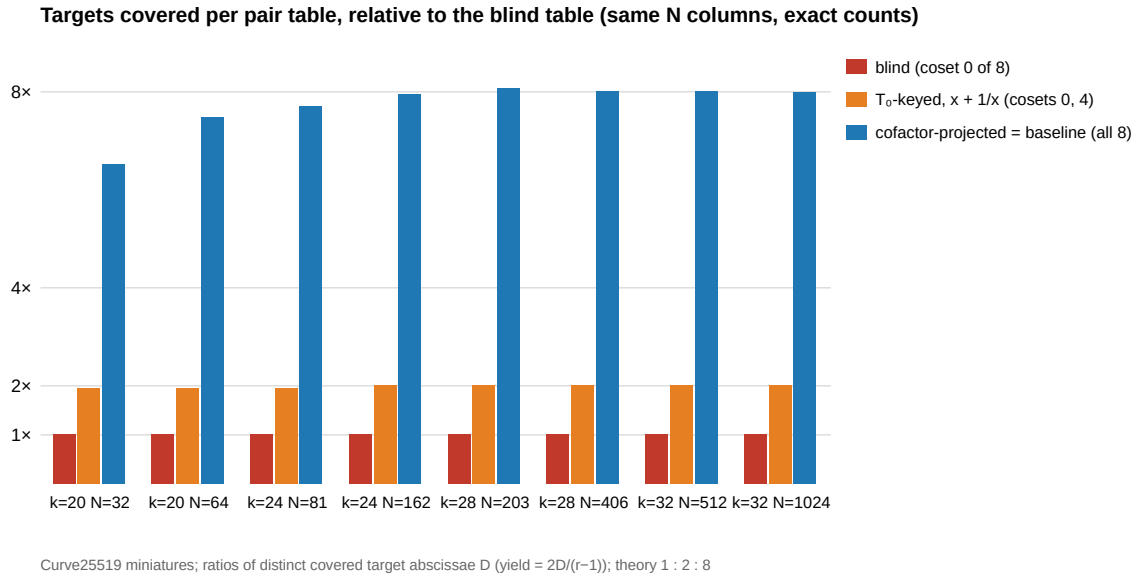
T_0 symmetry in Curve25519-family relation collection (2026-11-02)

Class: a mechanism measurement with exact counts. It is a negative result against the baseline and a 2× stage effect against the blind oracle; no pipeline improvement is claimed. Protocol: PROTOCOL.md, frozen before any count. Parent study: `../curve25519_ic_baseline_20261102`. PDF: REPORT.pdf.

What was tried

On $y^2 = x^3 + Ax^2 + x$, the rational 2-torsion point $T_0 = (0,0)$ acts on abscissae as $x \mapsto 1/x$, and $[8](P + T_0) = [8]P$, so P and $P + T_0$ have the same discrete log after the cofactor projection. A pair table keyed by the invariant $\kappa(S) = x(S) + 1/x(S)$ decomposes a subgroup target R when $R = \pm P_i \pm P_j$ **or** $R = \pm P_i \pm P_j + T_0$. The column count and the group operations stay the same, and each table entry and probe costs one extra (batchable) field inversion. `curve25519_family t0` builds the oracle, counts covered targets exactly over every pair, and runs it on 200,000 random targets per cell. Every hit is verified by point arithmetic.

Result



coverage

k	N	D blind	D T_0	D projected	T_0 ÷ blind	projected ÷ T_0	MC T_0 hits (verified)	z blind / T_0
20	32	154	301	1003	1.955	3.332	953 (953)	1.50 / 1.17
20	64	533	1037	3983	1.946	3.841	3124 (3124)	0.45 / −0.67
24	81	851	1654	6552	1.944	3.961	305 (305)	−0.42 / −0.59
24	162	3268	6549	25956	2.004	3.963	1195 (1195)	−1.25 / −1.53
28	203	5101	10286	41158	2.016	4.001	120 (120)	−0.62 / −0.24
28	406	20483	41090	163978	2.006	3.991	469 (469)	−0.33 / −0.94
32	512	32755	65593	262030	2.003	3.995	54 (54)	0.32 / 0.73

k	N	D blind	D T_0	D projected	T_0 ÷ blind	projected ÷ T_0	MC T_0 hits (verified)	z blind / T_0
32	1024	131021	261982	1046489	2.000	3.995	194 (194)	0.65 / −0.08

`D` is the number of distinct target abscissae covered, and yield is `2D/(r − 1)`. Raw output: analysis/t0.jsonl.

prediction	outcome
Q1: T_0 ÷ blind in [1.8, 2.2]	**holds**, 8/8 cells (1.944 to 2.016)
Q2: projected ÷ T_0 in [3.6, 4.4]	holds in 7/8 cells. **Misses** at k = 20, N = 32 (3.332): with 32 base points the coset labels are uneven (coset 0 holds 154 of 1,023 sums against 128 expected), so the blind count, and with it T_0's, sits above its mean. The ratio is at most 4 in every cell, so the conclusion stands.
Q3: every hit verified; MC within 4σ of exact	**holds**, 8/8 cells (all 6,414 T_0 hits verified, abs(z) ≤ 1.53)

Reading

- **The symmetry is real and the oracle works.** Keying by `x + 1/x` doubles the targets a blind table covers at no cost in columns or group operations. This is exactly the 2 of 8 cosets that $\langle T_0 \rangle$ reaches.
- **It cannot improve the baseline in the generic-decomposition track.** Translation by any torsion point can recover at most the cofactor, 8×. Cofactor projection, which the baseline already uses, recovers all 8, and T_0 alone recovers 2. Measured, projection beats T_0 by 3.8 to 4.0× at every size with N ≥ 64. Keying by `x([8]S)` instead, which is invariant under all of `E[8]`, is projection again with 3 extra doublings per probe. So no ecbench arm was run, as the protocol states.
- **Where it does matter: algebraic decomposition.** A Semaev/PKM system needs a factor base with an algebraic description, such as “small x”. Projection by 8 scrambles x, so that track cannot project and pays the full 8× cofactor penalty. There, closing the base under $x \mapsto 1/x$ gives the measured 2× in yield for free. It also gives the symmetry $S_3(1/x_1, 1/x_2, x_3) = S_3(x_1, x_2, x_3)/(x_1x_2)^2$ (unit-tested), which lets `S_{m+1}` be rewritten in T_0-invariant variables, the Faugère–Gaudry– Huot–Renault idea. The rest of the cofactor in that track would need the rational cyclic 8-isogeny $E \to E/\langle T_8 \rangle$, whose x-coordinate is the `E[8]`-invariant key. These are proposals, not measurements.

Next

1. On the algebraic track: a T_0-closed small-x base under `pkm-semaev` on the miniatures, against the same base unclosed. The predicted yield is 2×; the open question is the Gröbner cost of the symmetrised system.
2. The generic-track ratchet stays with the baseline’s candidates (base-size retune, closed fiber), which T_0 does not change.

Reproduce

```
        cargo build --release --bin curve25519_family
        for k in 20 24 28 32; do for n in <N_a> <2N_a>; do
          ./target/release/curve25519_family t0 $k $n 200000 $((2026110300+k)); done; done >
t0.jsonl
        ./target/release/curve25519_family t0-figure t0.jsonl t0.svg
```

`N_a` = 32, 81, 203, 512 for k = 20, 24, 28, 32. Run time is under 1 minute per cell.

PROTOCOL

T_0 symmetry in Curve25519-family relation collection — protocol

Frozen 2026-11-02 before any count was computed. Follows `../curve25519_ic_baseline_20261102` (baseline: cofactor-projected small-x base, `mitm:negation_folded=1`).

Question

On $E : y^2 = x^3 + Ax^2 + x$, translating by the 2-torsion point $T_0 = (0,0)$ acts as $x \mapsto 1/x$, and $[8](P + T_0) = [8]P$. Does exploiting it in an $m = 2$ pair-table decomposition raise relation yield? Does it improve on the baseline?

Mechanism and the bound (stated before measuring)

$E(F_p) \cong Z/8 \times Z/r$. A target R lies in the order-r subgroup G. Three oracles, all with N columns (one per base point up to sign), base points P_u at the smallest Montgomery abscissae $u \geq 2$:

- **blind**: table keyed by $x(P_i \pm P_j)$; it decomposes R only when a pair sum lies in G (coset 0 of 8).
- **t0**: table keyed by $\kappa(S) = x(S) + 1/x(S)$, which is invariant under $S \mapsto S + T_0$. A probe with $\kappa(R)$ decomposes R whenever a pair sum lies in G or $G + T_0$ (cosets 0 and 4), because $R = \pm P_i \pm P_j + T_0$ is a valid relation after the $[8]$-projection. Columns are unchanged.
- **proj** (the baseline's base): $P'_u = [8]P_u$; every pair sum lies in G.

The yield scales with the number of cosets covered: blind 1, t0 2, proj 8 (of 8). The ceiling for **any** torsion-translation symmetry is the full cofactor, which projection already attains. So:

- **Q1** (t0 vs blind): exact yield ratio in $[1.8, 2.2]$ at every size and N.
- **Q2** (proj vs t0): exact yield ratio in $[3.6, 4.4]$; t0 cannot beat the baseline in the generic-decomposition track.
- **Q3** (oracle works): Monte Carlo on random subgroup targets. Every t0 hit gives a relation verified by point arithmetic. The hit rate agrees with the exact yield within 4 binomial standard errors.

Falsification: Q1 fails outside $[1.8, 2.2]$. If Q2's ratio were below 1, the bound argument would be wrong and the t0 oracle would go to a full ecbench comparison against the baseline.

Measurement

`curve25519_family t0 <k> <N> <samples> <seed>` on the miniatures $k = 20, 24, 28, 32$ at $N \in \{N_a, 2N_a\}$ (as in the baseline). Exact: every pair $i \leq j$ with both signs. Each sum's coset comes from $c_i = [r]P_i \in E[8]$. D = distinct target abscissae covered, and $\text{yield} = 2D/(r - 1)$, as in `prime_small_x.rs`. Monte Carlo: 200,000 targets $[a]G$ per cell, seed $2026110300 + k$.

The cost side is stated rather than measured. Per probe, t0 adds one field inversion (batchable to about 3 multiplications) for $1/x(R)$; per table entry, the same. Group operations are equal to blind. So relations-phase cost per relation scales as 1/yield up to that field work.

What this is not

It is a mechanism measurement with exact counts, not an end-to-end ecbench S. If Q2 holds, no ecbench arm is warranted for the generic track.